

9.1.1 Aliasing: GATE CSE 2000 | Question: 1.16



Aliasing in the context of programming languages refers to

- A. multiple variables having the same memory location
- B. multiple variables having the same value
- C. multiple variables having the same identifier
- D. multiple uses of the same variable

gatecse-2000 programming easy aliasing

Answer key

9.2.1 Array: GATE CSE 2011 | Question: 22



What does the following fragment of C program print?

```
char c[] = "GATE2011";
char *p = c;
printf("%s", p + p[3] - p[1]);
```

- A. GATE2011
- B. E2011
- C. 2011
- D. 011

gatecse-2011 programming programming-in-c normal array

Answer key

9.2.2 Array: GATE CSE 2015 Set 1 | Question: 35



What is the output of the following C code? Assume that the address of x is 2000 (in decimal) and an integer requires four bytes of memory.

```
int main () {
    unsigned int x [4] [3] =
        {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}, {10, 11, 12}};
    printf ("%u, %u, %u", x + 3, *(x + 3), *(x + 2) + 3);
}
```

- A. 2036, 2036, 2036
- B. 2012, 4, 2204
- C. 2036, 10, 10
- D. 2012, 4, 6

gatecse-2015-set1 programming programming-in-c array normal

Answer key

9.2.3 Array: GATE CSE 2015 Set 3 | Question: 30



Consider the following two C code segments. Y and X are one and two dimensional arrays of size n and $n \times n$ respectively, where $2 \leq n \leq 10$. Assume that in both code segments, elements of Y are initialized to 0 and each element $X[i][j]$ of array X is initialized to $i + j$. Further assume that when stored in main memory all elements of X are in same main memory page frame.

Code segment 1 :

```
// initialize elements of Y to 0
// initialize elements of X[i][j] of X to i+j
for (i=0; i<n; i++)
    Y[i] += X[0][i];
```

Code segment 2 :

```
// initialize elements of Y to 0
// initialize elements of X[i][j] of X to i+j
for (i=0; i<n; i++)
    Y[i] += X[i][0];
```

Which of the following statements is/are correct?

S1: Final contents of array Y will be same in both code segments

S2: Elements of array X accessed inside the for loop shown in code segment 1 are contiguous in main memory

S3: Elements of array X accessed inside the for loop shown in code segment 2 are contiguous in main memory

A. Only S2 is correct

B. Only S3 is correct

C. Only S1 and S2 are correct

D. Only S1 and S3 are correct

gatecse-2015-set3 programming-in-c normal array

Answer key

9.2.4 Array: GATE CSE 2015 Set 3 | Question: 7



Consider the following C program segment.

```
#include <stdio.h>
int main()
{
    char s1[7] = "1234", *p;
    p = s1 + 2;
    *p = '0';
    printf("%s", s1);
}
```

What will be printed by the program?

A. 12

B. 120400

C. 1204

D. 1034

gatecse-2015-set3 programming programming-in-c normal array

Answer key

9.2.5 Array: GATE CSE 2019 | Question: 24



Consider the following C program:

```
#include <stdio.h>
int main() {
    int arr[]={1, 2, 3, 4, 5, 6, 7, 8, 9, 0, 1, 2, 5}, *ip=arr+4;
    printf("%d\n", ip[1]);
    return 0;
}
```

The number that will be displayed on execution of the program is _____

gatecse-2019 numerical-answers programming-in-c programming array easy one-mark

Answer key

9.2.6 Array: GATE CSE 2020 | Question: 22



Consider the following C program.

```
#include <stdio.h>
int main () {
    int a[4][5] = {{1, 2, 3, 4, 5},
                  {6, 7, 8, 9, 10},
                  {11, 12, 13, 14, 15},
                  {16, 17, 18, 19, 20}};
    printf("%d\n", *((a+**a+2)+3));
    return(0);
}
```

The output of the program is _____.

gatecse-2020 numerical-answers programming-in-c array one-mark

Answer key

9.2.7 Array: GATE IT 2004 | Question: 58



Consider the following C program which is supposed to compute the transpose of a given 4×4 matrix M . Note that, there is an X in the program which indicates some missing statements. Choose the correct option

to replace X in the program.

```
#include<stdio.h>
#define ROW 4
#define COL 4
int M[ROW][COL] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16};
main()
{
    int i, j, t;
    for (i = 0; i < 4; ++i)
    {
        X
    }
    for (i = 0; i < 4; ++i)
        for (j = 0; j < 4; ++j)
            printf ("%d", M[i][j]);
}
```

- A.

```
for(j = 0; j < 4; ++j){
    t = M[i][j];
    M[i][j] = M[j][i];
    M[j][i] = t;
}
```
- B.

```
for(j = 0; j < 4; ++j){
    M[i][j] = t;
    t = M[j][i];
    M[j][i] = M[i][j];
}
```
- C.

```
for(j = i; j < 4; ++j){
    t = M[i][j];
    M[i][j] = M[j][i];
    M[j][i] = t;
}
```
- D.

```
for(j = i; j < 4; ++j){
    M[i][j] = t;
    t = M[j][i];
    M[j][i] = M[i][j];
}
```

gateit-2004 programming easy programming-in-c array

Answer key

9.2.8 Array: GATE IT 2008 | Question: 49



What is the output printed by the following C code?

```
# include <stdio.h>
int main ()
{
    char a [6] = "world";
    int i, j;
    for (i = 0, j = 5; i < j; a [i++] = a [j--]);
    printf ("%s\n", a);
}
```

- A. dlrow B. Null string C. dlrlld D. worow

gateit-2008 programming programming-in-c normal array

Answer key

9.2.9 Array: GATE IT 2008 | Question: 51



Consider the C program given below. What does it print?

```
#include <stdio.h>
int main ()
{
    int i, j;
    int a [8] = {1, 2, 3, 4, 5, 6, 7, 8};
    for(i = 0; i < 3; i++) {
        a[i] = a[i] + 1;
        i++;
    }
    i--;
    for (j = 7; j > 4; j--) {
        int i = j/2;
        a[i] = a[i] - 1;
    }
    printf ("%d, %d", i, a[i]);
}
```

- A. 2,3 B. 2,4 C. 3,2 D. 3,3

Answer key

9.2.10 Array: GATE IT 2008 | Question: 52



C program is given below:

```
#include <stdio.h>
int main ()
{
    int i, j;
    char a [2] [3] = {{ 'a', 'b', 'c' }, { 'd', 'e', 'f' }};
    char b [3] [2];
    char *p = *b;
    for (i = 0; i < 2; i++) {
        for (j = 0; j < 3; j++) {
            *(p + 2*j + i) = a [i] [j];
        }
    }
}
```

What should be the contents of the array b at the end of the program?

- A. a b
c d
e f
- B. a d
b e
c f
- C. a c
e b
d f
- D. a e
d c
b f

Answer key

9.3

Functions (1)

9.3.1 Functions: GATE CSE 2024 | Set 2 | Question: 3



Consider the following C program. Assume parameters to a function are evaluated from *right to left*.

```
#include <stdio.h>

int g(int p) { printf("%d", p); return p; }

int h(int q) { printf("%d", q); return q; }

void f(int x, int y) {
    g(x);
    h(y);
}

int main() {
    f(g(10), h(20));
}
```

Which one of the following options is the CORRECT output of the above C program?

A. 20101020

B. 10202010

C. 20102010

D. 10201020

gatecse2024-set2 programming programming-in-c functions one-mark

Answer key

9.4

Goto (2)

9.4.1 Goto: GATE CSE 1989 | Question: 3-i



An unrestricted use of the "go to" statement is harmful because of which of the following reason (s):

- A. It makes it more difficult to verify programs.
- B. It makes programs more inefficient.
- C. It makes it more difficult to modify existing programs.
- D. It results in the compiler generating longer machine code.

gate1989 normal programming goto

Answer key

9.4.2 Goto: GATE CSE 1994 | Question: 1.5



An unrestricted use of the "goto" statement is harmful because

- A. it makes it more difficult to verify programs
- B. it increases the running time of the programs
- C. it increases the memory required for the programs
- D. it results in the compiler generating longer machine code

gate1994 programming easy goto

Answer key

9.5

Identify Function (6)

9.5.1 Identify Function: GATE CSE 1995 | Question: 3



Consider the following high level programming segment. Give the contents of the memory locations for variables W , X , Y and Z after the execution of the program segment. The values of the variables A and B are $5CH$ and $92H$, respectively. Also indicate error conditions if any.

```
var
  A, B, W, X, Y : unsigned byte;
  Z             : unsigned integer, (each integer is represented by two bytes)
begin
  X             := A+B
  Y             := abs(A-B);
  W             := A-B
  Z             := A*B
end;
```

gate1995 programming identify-function descriptive

Answer key

9.5.2 Identify Function: GATE CSE 1998 | Question: 2.13



What is the result of the following program?

```
program side-effect (input, output);
var x, result: integer;
function f (var x: integer): integer;
begin
  x:=x+1; f:=x;
end
begin
  x:=5;
  result:=f(x)*f(x);
  writeln(result);
end
```

A. 5

B. 25

C. 36

D. 42

gate1998 programming normal identify-function

Answer key

9.5.3 Identify Function: GATE CSE 2017 Set 2 | Question: 14



Consider the following function implemented in C:

```
void printxy(int x, int y) {
    int *ptr;
    x=0;
    ptr=&x;
    y=*ptr;
    *ptr=1;
    printf("%d, %d", x, y);
}
```

The output of invoking *printxy*(1, 1) is:

A. 0,0

B. 0,1

C. 1,0

D. 1,1

gatecse-2017-set2 programming-in-c identify-function pointers

Answer key

9.5.4 Identify Function: GATE CSE 2017 Set 2 | Question: 43



Consider the following snippet of a C program. Assume that *swap* (*&x*, *&y*) exchanges the content of *x* and *y*:

```
int main () {
    int array[] = {3, 5, 1, 4, 6, 2};
    int done =0;
    int i;
    while (done==0) {
        done =1;
        for (i=0; i<=4; i++) {
            if (array[i] < array[i+1]) {
                swap(&array[i], &array[i+1]);
                done=0;
            }
        }
        for (i=5; i>=1; i--) {
            if (array[i] > array[i-1]) {
                swap(&array[i], &array[i-1]);
                done =0;
            }
        }
    }
    printf("%d", array[3]);
}
```

The output of the program is _____

gatecse-2017-set2 programming algorithms numerical-answers identify-function

Answer key

9.5.5 Identify Function: GATE CSE 2019 | Question: 18



Consider the following C program :

```
#include<stdio.h>
int jumble(int x, int y){
    x = 2*x+y;
    return x;
}
int main(){
    int x=2, y=5;
    y=jumble(y,x);
    x=jumble(y,x);
    printf("%d \n",x);
    return 0;
}
```

The value printed by the program is _____.

gatecse-2019 programming-in-c numerical-answers identify-function one-mark

Answer key

9.5.6 Identify Function: GATE IT 2004 | Question: 15



Let x be an integer which can take a value of 0 or 1. The statement

```
if (x == 0) x = 1; else x = 0;
```

is equivalent to which one of the following ?

- A. $x = 1 + x$; B. $x = 1 - x$; C. $x = x - 1$; D. $x = 1 \% x$;

gateit-2004 programming easy identify-function

Answer key

9.6 Loop Invariants (8)

9.6.1 Loop Invariants: GATE CSE 1987 | Question: 7a



List the invariant assertions at points A, B, C, D and E in program given below:

```
Program division (input, output)
Const
  dividend = 81;
  divisor = 9;
Var remainder, quotient: interger
begin
  (*dividend >= 0) AND (divisor > 0)*)
  remainder := dividend;
  quotient := 0;
  (*A*)
  While (remainder >= 0) do
  begin (*B*)
    quotient := quotient + 1;
    remainder := remainder - divisor;
    (*C*)
  end;
  (*D*)
  quotient := quotient - 1;
  remainder := remainder + divisor;
  (*E*)
end
```

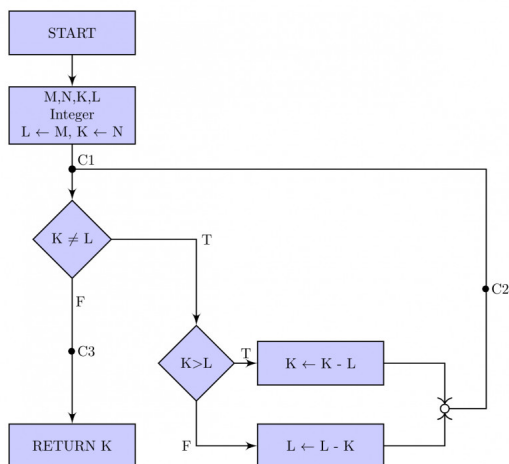
gate1987 programming loop-invariants descriptive

Answer key

9.6.2 Loop Invariants: GATE CSE 1988 | Question: 6ii



Below figure is the flow-chart corresponding to a program to calculate the gcd of two integers, M and N respectively, ($M, N > 0$). Use assertions at the cut point C_1, C_2 and C_3 to prove that the flow-chart is correct.



gate1988 normal descriptive loop-invariants

Answer key

9.6.3 Loop Invariants: GATE CSE 1988 | Question: 8ii



Consider the two program segments below:

a.

```
for
  i:=1 to f(x) by 1 do
  S
end
```

b.

```
i:=1;
While i<=f(x) do
  S
  i:=i+1
end
```

Under what conditions are these two programs equivalent? Treat S as any sequence of statements and f as a function.

gate1988 programming descriptive loop-invariants

Answer key

9.6.4 Loop Invariants: GATE CSE 1991 | Question: 1,vi



Consider the following PASCAL program segment:

```

if i mod 2 = 0 then
  while i >= 0 do
  begin
    i := i div 2;
    if i mod 2 < > 0 then i := i - 1;
    else i := i - 2;
  end;

```

An appropriate loop-invariant for the while-loop is _____

gate1991 programming loop-invariants normal fill-in-the-blanks

Answer key

9.6.5 Loop Invariants: GATE CSE 2004 | Question: 32



Consider the following program fragment for reversing the digits in a given integer to obtain a new integer.

Let $n = d_1 d_2 \dots d_m$.

```

int n, rev;
rev = 0;
while(n > 0) {
  rev = rev * 10 + n%10;

```

```

    n = n/10;
}

```

The loop invariant condition at the end of the i^{th} iteration is:

- A. $n = d_1 d_2 \dots d_{m-i}$ **and** $rev = d_m d_{m-1} \dots d_{m-i+1}$
- B. $n = d_{m-i+1} \dots d_{m-1} d_m$ **or** $rev = d_{m-i} \dots d_2 d_1$
- C. $n \neq rev$
- D. $n = d_1 d_2 \dots d_m$ **or** $rev = d_m \dots d_2 d_1$

gatecse-2004 programming loop-invariants normal

Answer key

9.6.6 Loop Invariants: GATE CSE 2015 Set 1 | Question: 33



Consider the following pseudo code, where x and y are positive integers.

```

begin
  q := 0
  r := x
  while r ≥ y do
    begin
      r := r - y
      q := q + 1
    end
  end
end

```

The post condition that needs to be satisfied after the program terminates is

- A. $\{r = qx + y \wedge r < y\}$
- B. $\{x = qy + r \wedge r < y\}$
- C. $\{y = qx + r \wedge 0 < r < y\}$
- D. $\{q + 1 < r - y \wedge y > 0\}$

gatecse-2015-set1 programming loop-invariants normal

Answer key

9.6.7 Loop Invariants: GATE CSE 2016 Set 2 | Question: 35



The following function computes X^Y for positive integers X and Y .

```

int exp (int X, int Y) {
  int res = 1, a = X, b = Y;

  while (b != 0) {
    if (b % 2 == 0) {a = a * a; b = b/2;}
    else {res = res * a; b = b - 1;}
  }
  return res;
}

```

Which one of the following conditions is TRUE before every iteration of the loop?

- A. $X^Y = a^b$
- B. $(res * a)^Y = (res * X)^b$
- C. $X^Y = res * a^b$
- D. $X^Y = (res * a)^b$

gatecse-2016-set2 programming loop-invariants normal

Answer key

9.6.8 Loop Invariants: GATE CSE 2017 Set 2 | Question: 37



Consider the C program fragment below which is meant to divide x by y using repeated subtractions. The variables x , y , q and r are all unsigned int.

```

while (r >= y) {
  r = r - y;
  q = q + 1;
}

```

Which of the following conditions on the variables x, y, q and r before the execution of the fragment will ensure that the loop terminated in a state satisfying the condition $x == (y * q + r)$?

- A. $(q == r) \ \&\& \ (r == 0)$
- B. $(x > 0) \ \&\& \ (r == x) \ \&\& \ (y > 0)$
- C. $(q == 0) \ \&\& \ (r == x) \ \&\& \ (y > 0)$
- D. $(q == 0) \ \&\& \ (y > 0)$

gatecse-2017-set2 programming loop-invariants

Answer key

9.7 Output (10)

9.7.1 Output: GATE CSE 2021 Set 2 | Question: 10



Consider the following ANSI C program.

```
#include <stdio.h>
int main()
{
    int arr[4][5];
    int i, j;
    for (i=0; i<4; i++)
    {
        for (j=0; j<5; j++)
        {
            arr[i][j] = 10 * i + j;
        }
    }
    printf("%d", *(arr[1]+9));
    return 0;
}
```

What is the output of the above program?

- A. 14
- B. 20
- C. 24
- D. 30

gatecse-2021-set2 programming-in-c array output one-mark

Answer key

9.7.2 Output: GATE CSE 2022 | Question: 33



What is printed by the following ANSI C program?

```
#include <stdio.h>
int main (int argc, char *argv[])
{
    int a[3][3][3] =
        {{1, 2, 3, 4, 5, 6, 7, 8, 9},
         {10, 11, 12, 13, 14, 15, 16, 17, 18},
         {19, 20, 21, 22, 23, 24, 25, 26, 27}};
    int i = 0, j = 0, k = 0;
    for ( i = 0; i < 3; i ++ ) {
        for ( k = 0; k < 3; k ++ )
            printf("%d", a[i][j][k]);
        printf ("\n");
    }
    return 0;
}
```

- 1 2 3
A. 10 11 12
19 20 21
1 4 7
B. 10 13 16
19 22 25
1 2 3
C. 4 5 6
7 8 9
1 2 3
D. 13 14 15
25 26 27

gatecse-2022 programming programming-in-c array output two-marks

Answer key

9.7.3 Output: GATE CSE 2022 | Question: 34



What is printed by the following ANSI C program?

```
#include<stdio.h>

int main(int argc, char *argv[]) {
    char a = 'P';
    char b = 'x';
    char c = (a&b) + '*';
    char d = (a|b) - '-';
    char e = (a^b) + '+';
    printf("%c %c %c\n", c, d, e);
    return 0;
}
```

ASCII encoding for relevant characters is given below

A	B	C	...	Z		a	b	c	...	z
65	66	67	...	90		97	98	99	...	122

*	+	-
42	43	45

- A. z K S B. 122 75 83 C. * - + D. P x +

gatecse-2022 programming programming-in-c output two-marks

Answer key

9.7.4 Output: GATE CSE 2023 | Question: 25



The integer value printed by the ANSI-C program given below is _____

```
#include<stdio.h>

int funcp(){
    static int x = 1;
    x++;
    return x;
}
```

```
int main(){
    int x,y;
    x = funcp();
    y = funcp()+x;
    printf("%d\n", (x+y));
    return 0;
}
```

gatecse-2023 programming programming-in-c output numerical-answers one-mark

Answer key

9.7.5 Output: GATE CSE 2024 | Set 1 | Question: 8



Consider the following C program:

```
#include <stdio.h>

int main() {
    int a=6;
    int b = 0;
    while (a<10) {
        a = a / 12+1 ;
        a += b ;}
    printf ("%d", a);
    return 0 ; }
```

Which one of the following statements is CORRECT?

- A. The program prints 9 as output
- B. The program prints 10 as output
- C. The program gets stuck in an infinite loop
- D. The program prints 6 as output

gatecse2024-set1 programming programming-in-c output one-mark

Answer key

9.7.6 Output: GATE CSE 2024 | Set 1 | Question: 9



Consider the following C program:

```
#include <stdio.h>
void fX ();
int main(){
    fX();
    return 0 ;}

void fX () {
    char a;
    if ((a=getchar()) != '\n')
        fX();
    if (a != '\n')
        putchar (a); }
```

Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

- A. The program will not terminate
- B. The program will terminate with no output
- C. The program will terminate with 4321 as output
- D. The program will terminate with 1234 as output

gatecse2024-set1 programming programming-in-c output one-mark

Answer key

9.7.7 Output: GATE CSE 2024 | Set 2 | Question: 23



Consider the following C function definition.

```
int fX(char *a) {
    char *b = a;
    while (*b)
        b++;
    return b - a; }

```

Which of the following statements is/are TRUE?

- A. The function call `fX("a b c d")` will always return a value
- B. Assuming a character array `c` is declared as `char c[] = "abcd"` in `main()`, the function call `fX(c)` will always return a value
- C. The code of the function will not compile
- D. Assuming a character pointer `c` is declared as `char *c = "abcd"` in `main()`, the function call `fX(c)` will always return a value

gategse2024-set2 programming programming-in-c multiple-selects output one-mark

Answer key

9.7.8 Output: GATE CSE 2025 | Set 1 | Question: 53



Consider the following C program:

```
#include <stdio.h>
int gate(int n) {
    int d, t, newnum, turn;
    newnum = turn = 0; t = 1;
    while (n >= t) t *= 10;
    t /= 10;
    while (t > 0) {
        d = n/t;
        n = n%t;
        t /= 10;
        if (turn) newnum = 10*newnum + d;
        turn = (turn + 1) % 2;
    }
    return newnum;
}
int main() {
    printf("%d", gate(14362));
    return 0;
}

```

The value printed by the given C program is _____. (Answer in integer)

gategse2025-set1 programming-in-c output numerical-answers two-marks

Answer key

9.7.9 Output: GATE CSE 2025 | Set 2 | Question: 23



```
int x=126,y=105;
do {
    if(x>y) x=x-y;
    else y=y-x;
} while(x!=y);
printf("%d",x);

```

The output of the given C code segment is _____. (Answer in integer)

Answer key

9.7.10 Output: GATE CSE 2025 | Set 2 | Question: 53



Consider the following C program:

```
#include <stdio.h>

int g(int n) {
    return (n+10);
}

int f(int n) {
    return g(n*2);
}

int main() {
    int sum, n;
    sum=0;
    for (n=1; n<3; n++)
        sum += g(f(n));
    printf ("%d", sum);
    return 0;
}
```

The output of the given C program is _____. (Answer in integer)

Answer key

9.8

Parameter Passing (12)

9.8.1 Parameter Passing: GATE CSE 1992 | Question: 10b



Show the activation records and the display structure just after the procedures called at lines marked x and y have started their execution. Be sure to indicate which of the two procedures named A you are referring to.

```
Program Test;
  Procedure A;
    Procedure B;
      Procedure A;
        begin
          .....
        end A;
      begin
        y: A;
      end B;
    begin
      B;
    end A;
  begin
    x: A;
  end Test
```

Answer key

9.8.2 Parameter Passing: GATE CSE 1994 | Question: 1.20



In which of the following cases is it possible to obtain different results for call-by-reference and call-by-name parameter passing methods?

- | | |
|--|---|
| A. Passing a constant value as a parameter | B. Passing the address of an array as a parameter |
| C. Passing an array element as a parameter | D. Passing an array |

Answer key

9.8.3 Parameter Passing: GATE CSE 2001 | Question: 2.17 | UGCNET-AUG2016-III: 21



What is printed by the print statements in the program *P1* assuming call by reference parameter passing?

```
Program P1()
{
    x = 10;
    y = 3;
    func1(y,x,x);
    print x;
    print y;
}

func1(x,y,z)
{
    y = y + 4;
    z = x + y + z
}
```

- A. 10, 3 B. 31, 3 C. 27, 7 D. None of the above

gatecse-2001 programming-in-c parameter-passing normal ugcnetcse-aug2016-paper3

Answer key

9.8.4 Parameter Passing: GATE CSE 2003 | Question: 73



The following program fragment is written in a programming language that allows global variables and does not allow nested declarations of functions.

```
global int i=100, j=5;
void P(x) {
    int i=10;
    print(x+10);
    i=200;
    j=20;
    print (x);
}
main() {P(i+j);}
```

If the programming language uses static scoping and call by need parameter passing mechanism, the values printed by the above program are:

- A. 115, 220 B. 25, 220 C. 25, 15 D. 115, 105

gatecse-2003 compiler-design normal runtime-environment parameter-passing

Answer key

9.8.5 Parameter Passing: GATE CSE 2008 | Question: 60



What is printed by the following C program?

```
int f(int x, int *py, int **ppz)
{
    int y, z;
    **ppz += 1; z = **ppz; // corrected z = *ppz; to z = **ppz;
    *py += 2; y = *py;
    x += 3;
    return x+y+z;
}

void main()
{
    int c, *b, **a;
    c = 4; b = &c; a = &b;
    printf("%d", f(c, b, a));
}
```

- A. 18 B. 19 C. 21 D. 22

gatecse-2008 programming programming-in-c normal parameter-passing

Answer key

9.8.6 Parameter Passing: GATE CSE 2010 | Question: 11



What does the following program print?

```
#include<stdio.h>

void f(int *p, int *q) {
    p=q;
    *p=2;
}

int i=0, j=1;

int main() {
    f(&i, &j);
    printf("%d %d\n", i,j);
    return 0;
}
```

A. 2 2

B. 2 1

C. 0 1

D. 0 2

gatecse-2010 programming programming-in-c easy parameter-passing

Answer key

9.8.7 Parameter Passing: GATE CSE 2013 | Question: 42



What is the return value of $f(p, p)$, if the value of p is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f (int &x, int c) {
    c = c - 1;
    if (c==0) return 1;
    x = x + 1;
    return f(x,c) * x;
}
```

gatecse-2013 compiler-design normal marks-to-all numerical-answers parameter-passing runtime-environment

Answer key

9.8.8 Parameter Passing: GATE CSE 2016 Set 1 | Question: 15



Consider the following C program.

```
# include <stdio.h>
void mystery (int *ptrA, int *ptrB) {
    int *temp;
    temp = ptrB;
    ptrB = ptrA;
    ptrA = temp;
}
int main () {
    int a = 2016, b=0, c= 4, d = 42;
    mystery (&a, &b);
    if (a < c)
        mystery (&c, &a);
    mystery (&a, &d);
    printf("%d\n", a);
}
```

The output of the program is _____.

gatecse-2016-set1 programming-in-c easy numerical-answers parameter-passing

Answer key

9.8.9 Parameter Passing: GATE CSE 2016 Set 2 | Question: 12



The value printed by the following program is _____.

```
void f (int *p, int m) {
```

```

    m = m + 5;
    *p = *p + m;
    return;
}
void main () {
    int i=5, j=10;

    f (&i, j);
    printf ("%d", i+j);
}

```

gatecse-2016-set2 programming-in-c normal numerical-answers parameter-passing

Answer key

9.8.10 Parameter Passing: GATE CSE 2018 | Question: 29



```

#include<stdio.h>
void fun1(char* s1, char* s2){
    char* temp;
    temp = s1;
    s1 = s2;
    s2 = temp;
}
void fun2(char** s1, char** s2){
    char* temp;
    temp = *s1;
    *s1 = *s2;
    *s2 = temp;
}
int main(){
    char *str1="Hi", *str2 = "Bye";
    fun1(str1, str2); printf("%s %s", str1, str2);
    fun2(&str1, &str2); printf("%s %s", str1, str2);
    return 0;
}

```

The output of the program above is:

- A. Hi Bye Bye Hi
- B. Hi Bye Hi Bye
- C. Bye Hi Hi Bye
- D. Bye Hi Bye Hi

gatecse-2018 programming-in-c pointers parameter-passing normal programming two-marks

Answer key

9.8.11 Parameter Passing: GATE IT 2006 | Question: 50



Which one of the choices given below would be printed when the following program is executed?

```

#include <stdio.h>
void swap (int *x, int *y)
{
    static int *temp;
    temp = x;
    x = y;
    y = temp;
}
void printab ()
{
    static int i, a = -3, b = -6;
    i = 0;
    while (i <= 4)
    {
        if ((i++)%2 == 1) continue;
        a = a + i;
        b = b + i;
    }
    swap (&a, &b);
    printf("a = %d, b = %d\n", a, b);
}
main()
{
    printab();
    printab();
}

```

- A. $a = 0, b = 3$
 $a = 0, b = 3$
 C. $a = 3, b = 6$
 $a = 3, b = 6$

- B. $a = 3, b = 0$
 $a = 12, b = 9$
 D. $a = 6, b = 3$
 $a = 15, b = 12$

gateit-2006 programming programming-in-c normal parameter-passing

Answer key

9.8.12 Parameter Passing: GATE IT 2008 | Question: 50



Consider the C program below. What does it print?

```
#include <stdio.h>
#define swap1(a, b) tmp = a; a = b; b = tmp
void swap2 (int a, int b)
{
    int tmp;
    tmp = a; a = b; b = tmp;
}
void swap3 (int*a, int*b)
{
    int tmp;
    tmp = *a; *a = *b; *b = tmp;
}
int main ()
{
    int num1 = 5, num2 = 4, tmp;
    if (num1 < num2) {swap1 (num1, num2);}
    if (num1 < num2) {swap2 (num1 + 1, num2);}
    if (num1 >= num2) {swap3 (&num1, &num2);}
    printf ("%d, %d", num1, num2);
}
```

- A. 5,5 B. 5,4 C. 4,5 D. 4,4

gateit-2008 programming programming-in-c easy parameter-passing

Answer key

9.9 Pointers (14)

9.9.1 Pointers: GATE CSE 2000 | Question: 1.12



The most appropriate matching for the following pairs

X : <code>m = malloc(5); m = NULL;</code>	1 : using dangling pointers
Y : <code>free(n); n->value = 5;</code>	2 : using uninitialized pointers
Z : <code>char *p, *p = 'a';</code>	3 : lost memory

is:

- A. $X - 1$ $Y - 3$ $Z - 2$ B. $X - 2$ $Y - 1$ $Z - 3$
 C. $X - 3$ $Y - 2$ $Z - 1$ D. $X - 3$ $Y - 1$ $Z - 2$

gatecse-2000 programming programming-in-c easy match-the-following pointers

Answer key

9.9.2 Pointers: GATE CSE 2001 | Question: 2.18



Consider the following three C functions:

[P1]

```
int *g(void)
{
    int x = 10;
    return (&x);
}
```

[P2]

```
int *g(void)
{
```

```
int *px;
*px = 10;
return px;
}
```

[P3]

```
int *g(void)
{
    int *px;
    px = (int*) malloc (sizeof(int));
    *px = 10;
    return px;
}
```

Which of the above three functions are likely to cause problems with pointers?

- A. Only *P3* B. Only *P1* and *P3* C. Only *P1* and *P2* D. *P1, P2* and *P3*

gatecse-2001 programming programming-in-c normal pointers

Answer key

9.9.3 Pointers: GATE CSE 2003 | Question: 2



Assume the following C variable declaration:

```
int *A[10], B[10][10];
```

Of the following expressions:

- I. $A[2]$
- II. $A[2][3]$
- III. $B[1]$
- IV. $B[2][3]$

which will not give compile-time errors if used as left hand sides of assignment statements in a C program?

- A. I, II, and IV only B. II, III, and IV only C. II and IV only D. IV only

gatecse-2003 programming programming-in-c easy pointers

Answer key

9.9.4 Pointers: GATE CSE 2003 | Question: 89



Consider the C program shown below:

```
#include<stdio.h>
#define print(x) printf("%d", x)

int x;
void Q(int z)
{
    z+=x;
    print(z);
}

void P(int *y)
{
    int x = *y + 2;
    Q(x);
    *y = x - 1;
    print(x);
}

main(void) {
    x = 5;
    P(&x);
    print(x);
}
```

The output of this program is:

- A. 12 7 6 B. 22 12 11 C. 14 6 6 D. 7 6 6

Answer key

9.9.5 Pointers: GATE CSE 2005 | Question: 1, ISRO2017-55



What does the following C-statement declare?

```
int (*f) (int * );
```

- A. A function that takes an integer pointer as argument and returns an integer
- B. A function that takes an integer as argument and returns an integer pointer
- C. A pointer to a function that takes an integer pointer as argument and returns an integer
- D. A function that takes an integer pointer as argument and returns a function pointer

Answer key

9.9.6 Pointers: GATE CSE 2006 | Question: 57



Consider this C code to swap two integers and these five statements: the code

```
void swap (int *px, int *py)
{
    *px = *px - *py;
    *py = *px + *py;
    *px = *py - *px;
}
```

- S1: will generate a compilation error
 S2: may generate a segmentation fault at runtime depending on the arguments passed
 S3: correctly implements the swap procedure for all input pointers referring to integers stored in memory locations accessible to the process
 S4: implements the swap procedure correctly for some but not all valid input pointers
 S5: may add or subtract integers and pointers

- A. S1
- B. S2 and S3
- C. S2 and S4
- D. S2 and S5

Answer key

9.9.7 Pointers: GATE CSE 2014 Set 1 | Question: 10



Consider the following program in C language:

```
#include <stdio.h>

main()
{
    int i;
    int*pi = &i;

    scanf("%d",pi);
    printf("%d\n", i+5);
}
```

Which one of the following statements is **TRUE**?

- A. Compilation fails.
- B. Execution results in a run-time error.
- C. On execution, the value printed is 5 more than the address of variable *i*.
- D. On execution, the value printed is 5 more than the integer value entered.

Answer key

9.9.8 Pointers: GATE CSE 2015 Set 3 | Question: 26



Consider the following C program

```
#include<stdio.h>
int main() {
    static int a[] = {10, 20, 30, 40, 50};
    static int *p[] = {a, a+3, a+4, a+1, a+2};
    int **ptr = p;
    ptr++;
    printf("%d%d", ptr-p, **ptr);
}
```

The output of the program is _____.

gatecse-2015-set3 programming programming-in-c normal numerical-answers pointers

Answer key

9.9.9 Pointers: GATE CSE 2017 Set 1 | Question: 13



Consider the following C code:

```
#include<stdio.h>
int *assignval (int *x, int val) {
    *x = val;
    return x;
}

void main () {
    int *x = malloc(sizeof(int));
    if (NULL == x) return;
    x = assignval (x,0);
    if (x) {
        x = (int *)malloc(sizeof(int));
        if (NULL == x) return;
        x = assignval (x,10);
    }
    printf("%d\n", *x);
    free(x);
}
```

The code suffers from which one of the following problems:

- A. compiler error as the return of *malloc* is not typecast appropriately.
- B. compiler error because the comparison should be made as $x == \text{NULL}$ and not as shown.
- C. compiles successfully but execution may result in dangling pointer.
- D. compiles successfully but execution may result in memory leak.

gatecse-2017-set1 programming-in-c programming pointers

Answer key

9.9.10 Pointers: GATE CSE 2017 Set 2 | Question: 55



Consider the following C program.

```
#include<stdio.h>
#include<string.h>
int main() {
    char* c="GATECSIT2017";
    char* p=c;
    printf("%d", (int)strlen(c+2[p]-6[p]-1));
    return 0;
}
```

The output of the program is _____

gatecse-2017-set2 programming-in-c numerical-answers array pointers

Answer key

9.9.11 Pointers: GATE CSE 2022 | Question: 11



What is printed by the following ANSI C program?

```
#include<stdio.h>

int main(int argc, char *argv[])
{
    int x = 1, z[2] = {10, 11};
    int *p = NULL;
    p = &x;
    *p = 10;
    p = &z[1];
    *(&z[0] + 1) += 3;
    printf("%d, %d, %d\n", x, z[0], z[1]);
    return 0;
}
```

A. 1,10,11

B. 1,10,14

C. 10,14,11

D. 10,10,14

gatecse-2022 programming programming-in-c pointers output one-mark

Answer key

9.9.12 Pointers: GATE CSE 2024 | Set 2 | Question: 26



What is the output of the following C program?

```
#include <stdio.h>
int main() {
    double a[2]={20.0,25.0}, *p,*q;
    p=a;
    q=p+1;
    printf("%d,%d", (int) (q-p), (int) (*q- *p));
    return 0;}

```

A. 4,8

B. 1,5

C. 8,5

D. 1,8

gatecse2024-set2 programming programming-in-c pointers two-marks

Answer key

9.9.13 Pointers: GATE CSE 2025 | Set 1 | Question: 24



```
#include <stdio.h>
void foo(int *p, int x) {
    *p=x;
}
int main(){
    int *z;
    int a = 20, b = 25;
    z = &a;
    foo(z,b);
    printf("%d",a);
    return 0;
}
```

The output of the given C program is _____. (Answer in integer)

gatecse2025-set1 programming-in-c pointers output numerical-answers easy one-mark

Answer key

9.9.14 Pointers: GATE CSE 2025 | Set 2 | Question: 52



Consider the following C program:

```
#include <stdio.h>
int main(){
    int a;
    int arr[5] = {30,50,10};
    int *ptr;
    ptr = &arr[0] + 1;
    a = *ptr;
    (*ptr)++;
    ptr++;
    printf("%d", a + (*ptr) + arr[1]);
    return 0;
}
```

The output of the above program is _____. (Answer in integer)

gatecse2025-set2 programming-in-c pointers output numerical-answers two-marks

Answer key

9.10

Programming Constructs (1)

9.10.1 Programming Constructs: GATE CSE 1999 | Question: 2.5



Given the programming constructs

- assignment
- for loops where the loop parameter cannot be changed within the loop
- if-then-else
- forward go to
- arbitrary go to
- non-recursive procedure call
- recursive procedure/function call
- repeat loop,

which constructs will you not include in a programming language such that it should be possible to program the terminates (i.e., halting) function in the same programming language

- A. (ii), (iii), (iv) B. (v), (vii), (viii)
C. (vi), (vii), (viii) D. (iii), (vii), (viii)

gate1999 programming normal programming-constructs

Answer key

9.11

Programming In C (31)

9.11.1 Programming In C: GATE CSE 2000 | Question: 2.20



The value of j at the end of the execution of the following C program:

```
int incr (int i)
{
    static int count = 0;
    count = count + i;
    return (count);
}
main () {
    int i, j;
    for (i = 0; i <= 4; i++)
        j = incr (i);
}
```

is:

- A. 10 B. 4 C. 6 D. 7

gatecse-2000 programming programming-in-c easy

Answer key

9.11.2 Programming In C: GATE CSE 2002 | Question: 1.17



In the C language:

- A. At most one activation record exists between the current activation record and the activation record for the main
- B. The number of activation records between the current activation record and the activation records from the main depends on the actual function calling sequence.
- C. The visibility of global variables depends on the actual function calling sequence
- D. Recursion requires the activation record for the recursive function to be saved in a different stack before the recursive function can be called.

gatecse-2002 programming programming-in-c easy descriptive

Answer key

9.11.3 Programming In C: GATE CSE 2002 | Question: 2.18



The C language is:

- A. A context free language
- B. A context sensitive language
- C. A regular language
- D. Parsable fully only by a Turing machine

gatecse-2002 programming programming-in-c normal

Answer key

9.11.4 Programming In C: GATE CSE 2002 | Question: 2.8



Consider the following declaration of a two-dimensional array in C:

```
char a[100][100];
```

Assuming that the main memory is byte-addressable and that the array is stored starting from memory address 0, the address of `a[40][50]` is:

- A. 4040
- B. 4050
- C. 5040
- D. 5050

gatecse-2002 programming-in-c programming easy

Answer key

9.11.5 Programming In C: GATE CSE 2004 | Question: 33



Consider the following C program segment:

```
char p[20]; int i;  
char* s = "string";  
int length = strlen(s);  
for(i = 0; i < length; i++)  
    p[i] = s[length-i];  
printf("%s", p);
```

The output of the program is:

- A. gnirts
- B. string
- C. gnirt
- D. no output is printed

gatecse-2004 programming programming-in-c easy

Answer key

9.11.6 Programming In C: GATE CSE 2005 | Question: 32



Consider the following C program:

```
double foo(double); /* Line 1 */  
int main() {  
    double da, db;  
    //input da  
    db = foo(da);  
}
```

```
double foo (double a) {
    return a;
}
```

The above code compiled without any error or warning. If Line 1 is deleted, the above code will show:

- A. no compile warning or error
- B. some compiler-warnings not leading to unintended results
- C. some compiler-warnings due to type-mismatch eventually leading to unintended results
- D. compiler errors

gatecse-2005 programming programming-in-c compiler-design easy

Answer key

9.11.7 Programming In C: GATE CSE 2008 | Question: 18



Which combination of the integer variables x , y , and z makes the variable a get the value 4 in the following expression?

$$a = (x > y) ? ((x > z) ? x : z) : ((y > z) ? y : z)$$

- A. $x = 3, y = 4, z = 2$
- B. $x = 6, y = 5, z = 3$
- C. $x = 6, y = 3, z = 5$
- D. $x = 5, y = 4, z = 5$

gatecse-2008 programming programming-in-c easy

Answer key

9.11.8 Programming In C: GATE CSE 2008 | Question: 61



Choose the correct option to fill ?1 and ?2 so that the program below prints an input string in reverse order. Assume that the input string is terminated by a new line character.

```
void reverse(void)
{
    int c;
    if(?1) reverse();
    ?2
}
main()
{
    printf("Enter text");
    printf("\n");
    reverse();
    printf("\n");
}
```

- A. ?1 is $(getchar() != '\n')$
?2 is $getchar(c);$
- B. ?1 is $((c = getchar()) != '\n')$
?2 is $getchar(c);$
- C. ?1 is $(c != '\n')$
?2 is $putchar(c);$
- D. ?1 is $((c = getchar()) != '\n')$
?2 is $putchar(c);$

gatecse-2008 programming normal programming-in-c

Answer key

9.11.9 Programming In C: GATE CSE 2012 | Question: 48



Consider the following C code segment.

```
int a, b, c = 0;
void prtFun(void);
```

```

main()
{
    static int a = 1;    /* Line 1 */
    prtFun();
    a += 1;
    prtFun();
    printf("\n %d %d ", a, b);
}

void prtFun(void)
{
    static int a = 2;    /* Line 2 */
    int b = 1;
    a += ++b;
    printf("\n %d %d ", a, b);
}

```

What output will be generated by the given code segment?

- | | | | |
|--------|--------|--------|--------|
| A. 3 1 | B. 4 2 | C. 4 2 | D. 3 1 |
| 4 1 | 6 1 | 6 2 | 5 2 |
| 4 2 | 6 1 | 2 0 | 5 2 |

gatecse-2012 programming programming-in-c normal

Answer key

9.11.10 Programming In C: GATE CSE 2012 | Question: 49



Consider the following C code segment.

```

int a, b, c = 0;
void prtFun(void);
main()
{
    static int a = 1;    /* Line 1 */
    prtFun();
    a += 1;
    prtFun();
    printf("\n %d %d ", a, b);
}

void prtFun(void)
{
    static int a = 2;    /* Line 2 */
    int b = 1;
    a += ++b;
    printf("\n %d %d ", a, b);
}

```

What output will be generated by the given code segment if:

Line 1 is replaced by **auto int a = 1;**

Line 2 is replaced by **register int a = 2;**

- | | | | |
|--------|--------|--------|--------|
| A. 3 1 | B. 4 2 | C. 4 2 | D. 4 2 |
| 4 1 | 6 1 | 6 2 | 4 2 |
| 4 2 | 6 1 | 2 0 | 2 0 |

normal gatecse-2012 programming-in-c programming

Answer key

9.11.11 Programming In C: GATE CSE 2014 Set 2 | Question: 11



Suppose n and p are unsigned int variables in a C program. We wish to set p to nC_3 . If n is large, which one of the following statements is most likely to set p correctly?

- | | |
|---|---|
| A. $p = n * (n - 1) * (n - 2) / 6;$ | B. $p = n * (n - 1) / 2 * (n - 2) / 3;$ |
| C. $p = n * (n - 1) / 3 * (n - 2) / 2;$ | D. $p = n * (n - 1) * (n - 2) / 6.0;$ |

gatecse-2014-set2 programming programming-in-c normal

Answer key

9.11.12 Programming In C: GATE CSE 2014 Set 2 | Question: 42



Consider the C function given below.

```
int f(int j)
{
    static int i = 50;
    int k;
    if (i == j)
    {
        printf("something");
        k = f(i);
        return 0;
    }
    else return 0;
}
```

Which one of the following is **TRUE**?

- A. The function returns 0 for all values of j .
- B. The function prints the string **something** for all values of j .
- C. The function returns 0 when $j = 50$.
- D. The function will exhaust the runtime stack or run into an infinite loop when $j = 50$.

gatecse-2014-set2 programming programming-in-c

Answer key

9.11.13 Programming In C: GATE CSE 2015 Set 1 | Question: 11



The output of the following C program is _____.

```
void f1 ( int a, int b) {
    int c;
    c = a; a = b;
    b = c;
}
void f2 ( int *a, int *b) {
    int c;
    c = *a; *a = *b; *b = c;
}
int main () {
    int a = 4, b = 5, c = 6;
    f1 ( a, b);
    f2 (&b, &c);
    printf ("%d", c - a - b);
}
```

gatecse-2015-set1 programming programming-in-c easy numerical-answers

Answer key

9.11.14 Programming In C: GATE CSE 2015 Set 3 | Question: 54



Consider the following C program:

```
#include<stdio.h>
int f1(void);
int f2(void);
int f3(void);
int x=10;
int main()
{
    int x=1;
    x += f1() + f2() + f3() + f2();
    printf("%d", x);
    return 0;
}
int f1() { int x = 25; x++; return x;}
int f2() { static int x = 50; x++; return x;}
int f3() { x *= 10; return x;}
```

The output of the program is _____.

Answer key

9.11.15 Programming In C: GATE CSE 2016 Set 1 | Question: 12



Consider the following "C" program.

```
void f(int, short);
void main()
{
    int i = 100;
    short s = 12;
    short *p = &s;
    _____; // call to f()
}
```

Which one of the following expressions , when placed in the blank above, will NOT result in a type checking error?

- A. $f(s, *s)$ B. $i = f(i, s)$ C. $f(i, *s)$ D. $f(i, *p)$

Answer key

9.11.16 Programming In C: GATE CSE 2016 Set 1 | Question: 34



The following function computes the maximum value contained in an integer array $P[]$ of size n ($n \geq 1$).

```
int max (int *p, int n) {
    int a = 0, b = n-1;

    while ( _____ ) {
        if (p[a] <= p[b]) {a = a+1;}
        else {b = b-1;}
    }
    return p[a];
}
```

The missing loop condition is:

- A. $a \neq n$ B. $b \neq 0$
 C. $b > (a + 1)$ D. $b \neq a$

Answer key

9.11.17 Programming In C: GATE CSE 2017 Set 1 | Question: 53



Consider the following C program.

```
#include<stdio.h>
#include<string.h>

void printlength(char *s, char *t) {
    unsigned int c=0;
    int len = ((strlen(s) - strlen(t)) > c) ? strlen(s) : strlen(t);
    printf("%d\n", len);
}

void main() {
    char *x = "abc";
    char *y = "defgh";
    printlength(x,y);
}
```

Recall that *strlen* is defined in *string.h* as returning a value of type *size_t*, which is an unsigned int. The output of the program is _____ .

Answer key

9.11.18 Programming In C: GATE CSE 2017 Set 1 | Question: 55



The output of executing the following C program is _____ .

```
#include<stdio.h>

int total(int v) {
    static int count = 0;
    while(v) {
        count += v&1;
        v >>= 1;
    }
    return count;
}

void main() {
    static int x=0;
    int i=5;
    for(; i>0; i--) {
        x = x + total(i);
    }
    printf("%d\n", x);
}
```

gatecse-2017-set1 programming programming-in-c normal numerical-answers

Answer key

9.11.19 Programming In C: GATE CSE 2017 Set 2 | Question: 2



Match the following:

P. static char var ;	i. Sequence of memory locations to store addresses
Q. m = malloc(10); m=NULL ;	ii. A variable located in data section of memory
R. char *ptr[10] ;	iii. Request to allocate a CPU register to store data
S. register int varl;	iv. A lost memory which cannot be freed

A. P-ii; Q-iv; R-i; S-iii
C. P-ii; Q-iv; R-iii; S-i

B. P-ii; Q-i; R-iv; S-iii
D. P-iii; Q-iv; R-i; S-ii

gatecse-2017-set2 programming programming-in-c match-the-following

Answer key

9.11.20 Programming In C: GATE CSE 2017 Set 2 | Question: 54



Consider the following C program.

```
#include<stdio.h>
int main () {
    int m=10;
    int n, n1;
    n=++m;
    n1=m++;
    n--;
    --n1;
    n-=n1;
    printf("%d", n);
    return 0;
}
```

The output of the program is _____

gatecse-2017-set2 programming-in-c numerical-answers easy

Answer key

9.11.21 Programming In C: GATE CSE 2018 | Question: 32



Consider the following C code. Assume that unsigned long int type length is 64 bits.

```
unsigned long int fun(unsigned long int n) {
    unsigned long int i, j=0, sum = 0;
```

```

for( i=n; i>1; i=i/2) j++;
for( ; j>1; j=j/2) sum++;
return sum;
}

```

The value returned when we call fun with the input 2^{40} is:

- A. 4 B. 5 C. 6 D. 40

gatecse-2018 programming-in-c normal programming two-marks

Answer key 

9.11.22 Programming In C: GATE CSE 2019 | Question: 27



Consider the following C program:

```

#include <stdio.h>
int r() {
    static int num=7;
    return num--;
}
int main() {
    for (r();r();r())
        printf("%d",r());
    return 0;
}

```

Which one of the following values will be displayed on execution of the programs?

- A. 41 B. 52 C. 63 D. 630

gatecse-2019 programming-in-c programming two-marks

Answer key 

9.11.23 Programming In C: GATE CSE 2019 | Question: 52



Consider the following C program:

```

#include <stdio.h>
int main() {
    float sum = 0.0, j=1.0, i=2.0;
    while (i/j > 0.0625) {
        j=j+i;
        sum=sum+i/j;
        printf("%f\n", sum);
    }
    return 0;
}

```

The number of times the variable sum will be printed, when the above program is executed, is _____

gatecse-2019 numerical-answers programming-in-c programming two-marks

Answer key 

9.11.24 Programming In C: GATE CSE 2019 | Question: 53



Consider the following C program:

```

#include <stdio.h>
int main()
{
    int a[] = {2, 4, 6, 8, 10};
    int i, sum=0, *b=a+4;
    for (i=0; i<5; i++)
        sum=sum+(*b-i)-*(b-i);
    printf("%d\n", sum);
    return 0;
}

```

The output of the above C program is _____

gatecse-2019 numerical-answers programming-in-c programming two-marks

9.11.25 Programming In C: GATE CSE 2021 Set 1 | Question: 37



Consider the following ANSI C program.

```
#include <stdio.h>
int main()
{
    int i, j, count;
    count=0;
    i=0;
    for (j=-3; j<=3; j++)
    {
        if ((j >= 0) && (i++))
            count = count + j;
    }
    count = count + i;
    printf("%d", count);
    return 0;
}
```

Which one of the following options is correct?

- A. The program will not compile successfully
- B. The program will compile successfully and output 10 when executed
- C. The program will compile successfully and output 8 when executed
- D. The program will compile successfully and output 13 when executed

gatecse-2021-set1 programming-in-c two-marks

9.11.26 Programming In C: GATE CSE 2024 | Set 1 | Question: 38



Consider the following C function definition.

```
int f (int x, int y){
    for (int i=0 ; i<y ; i++ ) {
        x= x + x + y;
    }
    return x;
}
```

Which of the following statements is/are TRUE about the above function?

- A. If the inputs are $x = 20, y = 10$, then the return value is greater than 2^{20}
- B. If the inputs are $x = 20, y = 20$, then the return value is greater than 2^{20}
- C. If the inputs are $x = 20, y = 10$, then the return value is less than 2^{10}
- D. If the inputs are $x = 10, y = 20$, then the return value is greater than 2^{20}

gatecse2024-set1 multiple-selects programming programming-in-c two-marks

9.11.27 Programming In C: GATE IT 2004 | Question: 59



What is the output of the following program?

```
#include<stdio.h>
int funcf (int x);
int funcg (int y);
main ()
{
    int x = 5, y = 10, count;
    for (count = 1; count <= 2; ++count) {
        y += funcf(x) + funcg(x);
        printf ("%d", y);
    }
}
funcf (int x) {
    int y;
```

```

    y = funcg(x);
    return (y);
}
funcg (int x) {
    static int y = 10;
    y += 1;
    return (y + x);
}

```

A. 43 80

B. 42 74

C. 33 37

D. 32 32

gateit-2004 programming programming-in-c normal

Answer key

9.11.28 Programming In C: GATE IT 2004 | Question: 60



Choose the correct option to fill the ?1 and ?2 so that the program prints an input string in reverse order. Assume that the input string is terminated by a new line character.

```

#include <stdio.h>
void wrt_it (void);
int main (void)
{
    printf("Enter Text");
    printf ("\n");
    wrt_it();
    printf ("\n");
    return 0;
}
void wrt_it (void)
{
    int c;
    if (?1)
        wrt_it();
    ?2
}

```

- A. ?1 is `getchar() != '\n'`
?2 is `getchar(c);`
- B. ?1 is `(c = getchar()); != '\n'`
?2 is `getchar(c);`
- C. ?1 is `c != '\n'`
?2 is `putchar(c);`
- D. ?1 is `(c = getchar()) != '\n'`
?2 is `putchar(c);`

gateit-2004 programming programming-in-c normal

Answer key

9.11.29 Programming In C: GATE IT 2005 | Question: 58



Let a be an array containing n integers in increasing order. The following algorithm determines whether there are two distinct numbers in the array whose difference is a specified number $S > 0$.

```

i = 0; j = 1;
while (j < n){
    if (E) j++;
    else if (a[j] - a[i] == S) break;
    else i++;
}
if (j < n) printf("yes") else printf ("no");

```

Choose the correct expression for E.

- A. $a[j] - a[i] > S$
- B. $a[j] - a[i] < S$
- C. $a[i] - a[j] < S$
- D. $a[i] - a[j] > S$

gateit-2005 programming normal programming-in-c

Answer key

9.11.30 Programming In C: GATE IT 2006 | Question: 51



Which one of the choices given below would be printed when the following program is executed?

```
#include <stdio.h>
int a1[] = {6, 7, 8, 18, 34, 67};
int a2[] = {23, 56, 28, 29};
int a3[] = {-12, 27, -31};
int *x[] = {a1, a2, a3};
void print(int *a[])
{
    printf("%d,", a[0][2]);
    printf("%d,", *a[2]);
    printf("%d,", **a[0]);
    printf("%d,", *(++a[0]);
    printf("%d\n", a[-1][+1]);
}
main()
{
    print(x);
}
```

- A. 8, -12, 7, 23, 8
C. -12, -12, 27, -31, 23

- B. 8, 8, 7, 23, 7
D. -12, -12, 27, -31, 56

gateit-2006 programming programming-in-c normal

Answer key

9.11.31 Programming In C: GATE IT 2007 | Question: 31



Consider the C program given below :

```
#include <stdio.h>
int main () {
    int sum = 0, maxsum = 0, i, n = 6;
    int a [] = {2, -2, -1, 3, 4, 2};
    for (i = 0; i < n; i++) {
        if (i == 0 || a[i] < 0 || a[i] < a[i - 1]) {
            if (sum > maxsum) maxsum = sum;
            sum = (a[i] > 0) ? a[i] : 0;
        }
        else sum += a[i];
    }
    if (sum > maxsum) maxsum = sum;
    printf ("%d\n", maxsum);
}
```

What is the value printed out when this program is executed?

- A. 9 B. 8 C. 7 D. 6

gateit-2007 programming programming-in-c normal

Answer key

9.12 Programming Paradigms (2)

9.12.1 Programming Paradigms: GATE CSE 2004 | Question: 1



The goal of structured programming is to:

- A. have well indented programs
B. be able to infer the flow of control from the compiled code
C. be able to infer the flow of control from the program text
D. avoid the use of GOTO statements

gatecse-2004 programming easy programming-paradigms

Answer key

9.12.2 Programming Paradigms: GATE CSE 2004 | Question: 90



Choose the best matching between the programming styles in Group 1 and their characteristics in Group 2.

Group 1	Group 2
P. Functional	1. Common-based, procedural
Q. Logic	2. Imperative, abstract data types
R. Object-oriented	3. Side-effect free, declarative, expression evaluations
S. Imperative	4. Declarative, clausal representation, theorem proving

A. $P-2$ $Q-3$ $R-4$ $S-1$
 C. $P-3$ $Q-4$ $R-1$ $S-2$

B. $P-4$ $Q-3$ $R-2$ $S-1$
 D. $P-3$ $Q-4$ $R-2$ $S-1$

gatecse-2004 programming normal programming-paradigms match-the-following

Answer key

9.13

Recursion (18)

9.13.1 Recursion: GATE CSE 1991 | Question: 01,x



Consider the following recursive definition of *fib*:

```
fib(n) := if n = 0 then 1
        else if n = 1 then 1
        else fib(n-1) + fib(n-2)
```

The number of times *fib* is called (including the first call) for evaluation of *fib*(7) is _____.

gate1991 programming recursion normal numerical-answers

Answer key

9.13.2 Recursion: GATE CSE 1994 | Question: 21



Consider the following recursive function:

```
function fib (n:integer):integer;
begin
if (n=0) or (n=1) then fib := 1
else fib := fib(n-1) + fib(n-2)
end;
```

The above function is run on a computer with a stack of 64 bytes. Assuming that only return address and parameter are passed on the stack, and that an integer value and an address takes 2 bytes each, estimate the maximum value of *n* for which the stack will not overflow. Give reasons for your answer.

gate1994 programming recursion normal descriptive

Answer key

9.13.3 Recursion: GATE CSE 2000 | Question: 16



A recursive program to compute Fibonacci numbers is shown below. Assume you are also given an array $f[0 \dots m]$ with all elements initialized to 0.

```
fib(n) {
  if (n > M) error ();
  if (n == 0) return 1;
  if (n == 1) return 1;
  if ( ) (1)
    return (2)
  t = fib(n - 1) + fib(n - 2);
  (3)
  return t;
}
```

- Fill in the boxes with expressions/statement to make *fib*() store and reuse computed Fibonacci values. Write the box number and the corresponding contents in your answer book.
- What is the time complexity of the resulting program when computing *fib*(*n*)?

Answer key

9.13.4 Recursion: GATE CSE 2001 | Question: 13



Consider the following C program:

```
void abc(char*s)
{
    if(s[0]=='\0')return;
    abc(s+1);
    printf("%c",s[0]);
}

main()
{
    abc("123");
}
```

- A. What will be the output of the program?
 B. If $abc(s)$ is called with a null-terminated string s of length n characters (not counting the null (' $\backslash 0$ ') character), how many characters will be printed by $abc(s)$?

Answer key

9.13.5 Recursion: GATE CSE 2002 | Question: 11



The following recursive function in C is a solution to the Towers of Hanoi problem.

```
void move(int n, char A, char B, char C) {
    if (.....) {
        move (.....);
        printf("Move disk %d from pole %c to pole %c\n", n, A, C);
        move (.....);
    }
}
```

Fill in the dotted parts of the solution.

Answer key

9.13.6 Recursion: GATE CSE 2004 | Question: 31, ISRO2008-40



Consider the following C function:

```
int f(int n)
{
    static int i = 1;
    if(n >= 5) return n;
    n = n+i;
    i++;
    return f(n);
}
```

The value returned by $f(1)$ is:

- A. 5 B. 6 C. 7 D. 8

Answer key

9.13.7 Recursion: GATE CSE 2005 | Question: 81b



```
double foo(int n)
{
```

```

int i;
double sum;
if(n == 0)
{
    return 1.0;
}
else
{
    sum = 0.0;
    for(i = 0; i < n; i++)
    {
        sum += foo(i);
    }
    return sum;
}
}

```

Suppose we modify the above function $foo()$ and stores the value of $foo(i)$ $0 \leq i < n$, as and when they are computed. With this modification the time complexity for function $foo()$ is significantly reduced. The space complexity of the modified function would be:

- A. $O(1)$ B. $O(n)$ C. $O(n^2)$ D. $n!$

gatecse-2005 programming recursion normal

Answer key

9.13.8 Recursion: GATE CSE 2007 | Question: 42



Consider the following C function:

```

int f(int n)
{
    static int r = 0;
    if (n <= 0) return 1;
    if (n > 3)
    {
        r = n;
        return f(n-2) + 2;
    }
    return f(n-1) + r;
}

```

What is the value of $f(5)$?

- A. 5 B. 7 C. 9 D. 18

gatecse-2007 programming recursion normal

Answer key

9.13.9 Recursion: GATE CSE 2014 Set 2 | Question: 40



Consider the following function.

```

double f(double x){
    if( abs(x*x - 3) < 0.01)
        return x;
    else
        return f(x/2 + 1.5/x);
}

```

Give a value q (to 2 decimals) such that $f(q)$ will return q :_____.

gatecse-2014-set2 programming recursion numerical-answers normal

Answer key

9.13.10 Recursion: GATE CSE 2015 Set 2 | Question: 15



Consider the following function written in the C programming language :

```

void foo(char *a)
{
    if (*a && *a != '\0')
    {

```

```

foo(a+1);
putchar(*a);
}

```

The output of the above function on input "*ABCD EFGH*" is

- A. *ABCD EFGH* B. *ABCD* C. *HGFE DCBA* D. *DCBA*

gatecse-2015-set2 programming programming-in-c normal recursion

Answer key 

9.13.11 Recursion: GATE CSE 2016 Set 1 | Question: 35



What will be the output of the following *C* program?

```

void count (int n) {
    static int d=1;

    printf ("%d",n);
    printf ("%d",d);
    d++;
    if (n>1) count (n-1);
    printf ("%d",d);
}

void main(){
    count (3);
}

```

- A. 3 1 2 2 1 3 4 4 4
 B. 3 1 2 1 1 1 2 2 2
 C. 3 1 2 2 1 3 4
 D. 3 1 2 1 1 1 2

gatecse-2016-set1 programming-in-c recursion normal

Answer key 

9.13.12 Recursion: GATE CSE 2016 Set 2 | Question: 37



Consider the following program:

```

int f (int * p, int n)
{
    if (n <= 1) return 0;
    else return max (f (p+1, n-1), p[0] - p[1]);
}

int main ()
{
    int a[] = {3, 5, 2, 6, 4};
    printf ("%d", f(a, 5));
}

```

Note: $\max(x, y)$ returns the maximum of x and y .

The value printed by this program is _____.

gatecse-2016-set2 programming-in-c normal numerical-answers recursion

Answer key 

9.13.13 Recursion: GATE CSE 2017 Set 1 | Question: 35



Consider the following two functions.

```

void fun1(int n) {
    if(n == 0) return;
    printf("%d", n);
    fun2(n - 2);
    printf("%d", n);
}

```

```
void fun2(int n) {
    if(n == 0) return;
    printf("%d", n);
    fun1(++n);
    printf("%d", n);
}
```

The output printed when `fun1(5)` is called is

- A. 53423122233445
- B. 53423120112233
- C. 53423122132435
- D. 53423120213243

gatecse-2017-set1 programming normal tricky recursion

Answer key

9.13.14 Recursion: GATE CSE 2017 Set 1 | Question: 36



Consider the C functions `foo` and `bar` given below:

```
int foo(int val) {
    int x=0;
    while(val > 0) {
        x = x + foo(val--);
    }
    return val;
}
```

```
int bar(int val) {
    int x = 0;
    while(val > 0) {
        x = x + bar(val-1);
    }
    return val;
}
```

Invocations of `foo(3)` and `bar(3)` will result in:

- A. Return of 6 and 6 respectively.
- B. Infinite loop and abnormal termination respectively.
- C. Abnormal termination and infinite loop respectively.
- D. Both terminating abnormally.

gatecse-2017-set1 programming-in-c programming normal recursion

Answer key

9.13.15 Recursion: GATE CSE 2018 | Question: 21



Consider the following C program:

```
#include<stdio.h>

int counter=0;

int calc (int a, int b) {
    int c;
    counter++;
    if(b==3) return (a*a*a);
    else {
        c = calc(a, b/3);
        return (c*c*c);
    }
}

int main() {
    calc(4, 81);
    printf("%d", counter);
}
```

The output of this program is _____.

gatecse-2018 programming-in-c numerical-answers recursion programming one-mark

Answer key

9.13.16 Recursion: GATE CSE 2020 | Question: 46



Consider the following C functions.

```
int fun1(int n) {
    static int i = 0;
    if (n > 0) {
        ++i;
        fun1(n-1);
    }
    return (i);
}
```

```
int fun2(int n) {
    static int i = 0;
    if (n > 0) {
        i = i + fun1(n);
        fun2(n-1);
    }
    return (i);
}
```

The return value of `fun2(5)` is _____

gatecse-2020 numerical-answers programming-in-c recursion two-marks

Answer key

9.13.17 Recursion: GATE CSE 2025 | Set 1 | Question: 51



```
#include <stdio.h>
int foo(int S[], int size) {
    if(size == 0) return 0;
    if(size == 1) return 1;
    if(S[0] != S[1]) return 1+foo(S+1, size-1);
    return foo(S+1, size-1);
}
int main() {
    int A[]={0,1,2,2,0,0,1,1};
    printf("%d", foo(A, 9));
    return 0;
}
```

The value printed by the given C program is _____. (Answer in integer)

gatecse2025-set1 programming-in-c recursion output numerical-answers two-marks

Answer key

9.13.18 Recursion: GATE IT 2007 | Question: 27



The function `f` is defined as follows:

```
int f (int n) {
    if (n <= 1) return 1;
    else if (n % 2 == 0) return f(n/2);
    else return f(3n - 1);
}
```

Assuming that arbitrarily large integers can be passed as a parameter to the function, consider the following statements.

- The function f terminates for finitely many different values of $n \geq 1$.
- The function f terminates for infinitely many different values of $n \geq 1$.
- The function f does not terminate for finitely many different values of $n \geq 1$.
- The function f does not terminate for infinitely many different values of $n \geq 1$.

Which one of the following options is true of the above?

- A. i and iii B. i and iv C. ii and iii D. ii and iv

gateit-2007 programming recursion normal

Answer key

9.14.1 Strings: GATE CSE 2025 | Set 2 | Question: 9



Consider the following C program:

```
#include <stdio.h>
void stringcopy (char *, char *);
int () {
    char a[30] "@Hello World";
    stringcopy(, a+2);
    printf("%s\n",a);
    return 0;
}
void stringcopy(char *s, char *t) {
    while (*t)
        *s++=*t++;
}
```

Which one of the following will be the output of the program?

- A. @#Hello World! B. Hello World! C. ello World! D. @#Hello World!d!

gatecse2025-set2 programming-in-c strings output one-mark

Answer key

9.15

Structure (5)

9.15.1 Structure: GATE CSE 2000 | Question: 1.11



The following C declarations:

```
struct node {
    int i;
    float j;
};
struct node *s[10];
```

define s to be:

- A. An array, each element of which is a pointer to a structure of type node
 B. A structure of 2 fields, each field being a pointer to an array of 10 elements
 C. A structure of 3 fields: an integer, a float, and an array of 10 elements
 D. An array, each element of which is a structure of type node

gatecse-2000 programming programming-in-c easy structure

Answer key

9.15.2 Structure: GATE CSE 2018 | Question: 2



Consider the following C program:

```
#include<stdio.h>
struct Ournode{
    char x, y, z;
};
int main() {
    struct Ournode p={'1', '0', 'a'+2};
    struct Ournode *q=&p;
    printf("%c, %c", *((char*)q+1), *((char*)q+2));
    return 0;
}
```

The output of this program is:

- A. 0, c B. 0, a+2 C. '0', 'a+2' D. '0', 'c'

gatecse-2018 programming-in-c programming structure normal one-mark

Answer key

9.15.3 Structure: GATE CSE 2021 Set 2 | Question: 35



Consider the following ANSI C program:

```
#include <stdio.h>
#include <stdlib.h>
struct Node{
    int value;
    struct Node *next;};
int main() {
    struct Node *boxE, *head, *boxN; int index=0;
    boxE=head= (struct Node *) malloc(sizeof(struct Node));
    head → value = index;
    for (index =1; index<=3; index++){
        boxN = (struct Node *) malloc (sizeof(struct Node));
        boxE → next = boxN;
        boxN → value = index;
        boxE = boxN; }
    for (index=0; index<=3; index++) {
        printf("Value at index %d is %d\n", index, head → value);
        head = head → next;
        printf("Value at index %d is %d\n", index+1, head → value); } }
```

Which one of the following statements below is correct about the program?

- A. Upon execution, the program creates a linked-list of five nodes
- B. Upon execution, the program goes into an infinite loop
- C. It has a missing **return** which will be reported as an error by the compiler
- D. It dereferences an uninitialized pointer that may result in a run-time error

gatecse-2021-set2 programming-in-c normal pointers structure two-marks

Answer key

9.15.4 Structure: GATE IT 2004 | Question: 61



Consider the following C program:

```
#include <stdio.h>
typedef struct {
    char *a;
    char *b;
} t;
void f1 (t s);
void f2 (t *p);
main()
{
    static t s = {"A", "B"};
    printf ("%s %s\n", s.a, s.b);
    f1(s);
    printf ("%s %s\n", s.a, s.b);
    f2(&s);
}
void f1 (t s)
{
    s.a = "U";
    s.b = "V";
    printf ("%s %s\n", s.a, s.b);
    return;
}
void f2(t *p)
{
    p -> a = "V";
    p -> b = "W";
    printf ("%s %s\n", p -> a, p -> b);
    return;
}
```

What is the output generated by the program ?

- | | |
|---------------|---------------|
| A. <i>A B</i> | B. <i>A B</i> |
| <i>U V</i> | <i>U V</i> |
| <i>V W</i> | <i>A B</i> |
| <i>V W</i> | <i>V W</i> |
| C. <i>A B</i> | D. <i>A B</i> |

U V
U V
V W

U V
V W
U V

gateit-2004 programming programming-in-c normal structure

Answer key

9.15.5 Structure: GATE IT 2006 | Question: 49



Which one of the choices given below would be printed when the following program is executed ?

```
#include <stdio.h>
struct test {
    int i;
    char *c;
}st[] = {5, "become", 4, "better", 6, "jungle", 8, "ancestor", 7, "brother"};
main ()
{
    struct test *p = st;
    p += 1;
    ++p -> c;
    printf("%s", p++ -> c);
    printf("%c", *++p -> c);
    printf("%d", p[0].i);
    printf("%s\n", p -> c);
}
```

- A. jungle, n, 8, nclastor
- C. cetter, k, 6, jungle

- B. etter, u, 6, ungle
- D. etter, u, 8, ncestor

gateit-2006 programming programming-in-c normal structure

Answer key

9.16 Switch Case (2)

9.16.1 Switch Case: GATE CSE 2012 | Question: 3



What will be the output of the following C program segment?

```
char inChar = 'A';
switch ( inChar ) {
    case 'A' : printf ("Choice A\n");
    case 'B' :
    case 'C' : printf ("Choice B");
    case 'D' :
    case 'E' :
    default : printf ("No Choice");
}
```

- A. No Choice
- C. Choice A
Choice B No Choice

- B. Choice A
- D. Program gives no output as it is erroneous

gatecse-2012 programming easy programming-in-c switch-case

Answer key

9.16.2 Switch Case: GATE CSE 2015 Set 3 | Question: 48



Consider the following C program:

```
#include<stdio.h>
int main()
{
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k=-j;
    for (i=0; i<5; i++)
    {
        switch(i+k)
        {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
}
```

```
} return 0;
```

The number of times printf statement is executed is _____.

gatecse-2015-set3 programming programming-in-c switch-case normal numerical-answers

Answer key

9.17 Type Checking (1)

9.17.1 Type Checking: GATE CSE 2003 | Question: 24



Which of the following statements is FALSE?

- A. In statically typed languages, each variable in a program has a fixed type
- B. In un-typed languages, values do not have any types
- C. In dynamically typed languages, variables have no types
- D. In all statically typed languages, each variable in a program is associated with values of only a single type during the execution of the program

gatecse-2003 programming normal type-checking

Answer key

9.18 Union (1)

9.18.1 Union: GATE CSE 2000 | Question: 1.17, ISRO2015-79



Consider the following C declaration:

```
struct {
    short x[5];
    union {
        float y;
        long z;
    } u;
}t;
```

Assume that the objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable *t*, ignoring alignment consideration, is:

- A. 22 bytes
- B. 14 bytes
- C. 18 bytes
- D. 10 bytes

gatecse-2000 programming programming-in-c easy isro2015 union

Answer key

9.19 Variable Binding (1)

9.19.1 Variable Binding: GATE IT 2007 | Question: 34, UGCNET-Dec2012-III: 52



Consider the program below in a hypothetical programming language which allows global variables and a choice of static or dynamic scoping.

```
int i;
program main()
{
    i = 10;
    call f();
}

procedure f()
{
    int i = 20;
    call g ();
}

procedure g()
{
    print i;
}
```

Let x be the value printed under static scoping and y be the value printed under dynamic scoping. Then, x and y are:

A. $x = 10, y = 20$

B. $x = 20, y = 10$

C. $x = 10, y = 10$

D. $x = 20, y = 20$

gateit-2007 programming variable-binding easy ugcnetcse-dec2012-paper3

Answer key

Answer Keys

9.1.1	A	9.2.1	C	9.2.2	A	9.2.3	C	9.2.4	C
9.2.5	6	9.2.6	19	9.2.7	C	9.2.8	B	9.2.9	C
9.2.10	B	9.3.1	A	9.4.1	A	9.4.2	A	9.5.1	N/A
9.5.2	C	9.5.3	C	9.5.4	3	9.5.5	26	9.5.6	B
9.6.1	N/A	9.6.2	N/A	9.6.3	N/A	9.6.4	N/A	9.6.5	A
9.6.6	B	9.6.7	C	9.6.8	C	9.7.1	C	9.7.2	A
9.7.3	A	9.7.4	7	9.7.5	C	9.7.6	C	9.7.7	A;B;D
9.7.8	46:46	9.7.9	21:21	9.7.10	46:46	9.8.1	N/A	9.8.2	C
9.8.3	B	9.8.4	D	9.8.5	B	9.8.6	D	9.8.7	6561
9.8.8	2016	9.8.9	30	9.8.10	A	9.8.11	D	9.8.12	C
9.9.1	D	9.9.2	C	9.9.3	A	9.9.4	A	9.9.5	C
9.9.6	C	9.9.7	D	9.9.8	140	9.9.9	D	9.9.10	2
9.9.11	D	9.9.12	B	9.9.13	25:25	9.9.14	111:111	9.10.1	B
9.11.1	A	9.11.2	B	9.11.3	B	9.11.4	B	9.11.5	D
9.11.6	D	9.11.7	A	9.11.8	D	9.11.9	C	9.11.10	D
9.11.11	B	9.11.12	D	9.11.13	-5	9.11.14	230	9.11.15	D
9.11.16	D	9.11.17	3	9.11.18	23	9.11.19	A	9.11.20	0
9.11.21	B	9.11.22	B	9.11.23	5	9.11.24	10	9.11.25	B
9.11.26	B;D	9.11.27	A	9.11.28	D	9.11.29	B	9.11.30	A
9.11.31	C	9.12.1	C	9.12.2	D	9.13.1	41	9.13.2	16
9.13.3	N/A	9.13.4	N/A	9.13.5	N/A	9.13.6	C	9.13.7	B
9.13.8	D	9.13.9	1.72 : 1.74	9.13.10	D	9.13.11	A	9.13.12	3
9.13.13	A	9.13.14	C	9.13.15	4	9.13.16	55	9.13.17	5:5
9.13.18	D	9.14.1	D	9.15.1	A	9.15.2	A	9.15.3	D
9.15.4	B	9.15.5	B	9.16.1	X	9.16.2	10	9.17.1	C
9.18.1	C	9.19.1	A						